

A. Ten patterns cover every output-one state

Fixed ones and free positions	111**	1	1	1	*	*
	11*1*	1	1	*	1	*
	11**1	1	1	*	*	1
	1*11*	1	*	1	1	*
	1*1*1	1	*	1	*	1
	1**11	1	*	*	1	1
	111	*	1	1	1	*
	*11*1	*	1	1	*	1
	*1*11	*	1	*	1	1
	**111	*	*	1	1	1
		x ₀	x ₁	x ₂	x ₃	x ₄

C. Unfold one pattern and test one input

Pattern 111** fixes three ones.

Anchor: $1 + 2 + 4 = 7$

Free positions: 3 and 4

Offsets: {0, 8, 16, 24}

Indices: {7, 15, 23, 31}

The four orange rows all have y = 1.

Flip only input 0 at the boundary:

01100 → 11100

index 6, y = 0 → index 7, y = 1

Input 0 matters to the function.

B. Expand into the complete repertoire

0	0	0	0	0	0	0
1	1	0	0	0	0	0
2	0	1	0	0	0	0
3	1	1	0	0	0	0
4	0	0	1	0	0	0
5	1	0	1	0	0	0
6	0	1	1	0	0	0
7	1	1	1	0	0	1
8	0	0	0	1	0	0
9	1	0	0	1	0	0
10	0	1	0	1	0	0
11	1	1	0	1	0	1
12	0	0	1	1	0	0
13	1	0	1	1	0	1
14	0	1	1	1	0	1
15	1	1	1	1	0	1
16	0	0	0	0	1	0
17	1	0	0	0	1	0
18	0	1	0	0	1	0
19	1	1	0	0	1	1
20	0	0	1	0	1	0
21	1	0	1	0	1	1
22	0	1	1	0	1	1
23	1	1	1	0	1	1
24	0	0	0	1	1	0
25	1	0	0	1	1	1
26	0	1	0	1	1	1
27	1	1	0	1	1	1
28	0	0	1	1	1	1
29	1	0	1	1	1	1
30	0	1	1	1	1	1
31	1	1	1	1	1	1
	x ₀	x ₁	x ₂	x ₃	x ₄	y